# HAAJAR SAEED ARCHER

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**Country of Residence:** Qatar | **Nationality:** Canadian | **Languages:** English (Native), Arabic (Conversational)

## EDUCATION

**BACHELOR OF SCIENCE (HONOURS) IN COMPUTER SCIENCE WITH A SPECIALISM IN DATA ANALYTICS** Sep 2022 – Dec 2025
Asia Pacific University of Technology & Innovation (APU) – Kuala Lumpur, Malaysia

## SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL
- **Web & Backend:** Next.js, React, Node.js, Convex, Clerk, FastAPI, Tailwind CSS, REST APIs & webhooks
- **AI & Agents:** LLM agents & tool calling, RAG & vector search, evals, Claude Code, Vercel AI SDK, Claude & OpenAI APIs, Composio, MCP
- **Data & ML:** Pandas, NumPy, Matplotlib, XGBoost, CatBoost, TensorFlow/Keras, BERT, SHAP, Machine & Deep Learning
- **Databases:** PostgreSQL, SQL Server, MongoDB, Redis
- **Analytics & BI:** PostHog, Power BI (DAX, Data Modelling, Dashboards), Google Analytics, SAS Enterprise Miner, R Studio, Excel
- **Tools & DevOps:** Git, GitHub Actions, Vercel, Docker, Vitest, Playwright, Sentry, Linear, Slack API, Word, PowerPoint
- **Certification:** CCNA: Introduction to Networks – Cisco

## WORK EXPERIENCE

### AGENTIC OPERATOR INTERN

Utopia Studio – Doha, Qatar Jun 2026 – Present

- Top engineering contributor to the studio's two internal AI platforms: **~1,800 commits (~75% of each codebase)** and **2,800+ automated tests**, built with TypeScript, Next.js, React, Convex, Clerk and Vercel.

***Utopia OS*** *— the studio operating system used by fellows, the studio team and admins*

- Built end-to-end **fellow onboarding** across the fellow and studio/admin workspaces: multi-step forms with draft autosave, Google Drive document uploads, Ashby auto-fill and Calendly booking, plus an admin onboarding desk to review submissions, unlock modules and manage the fellow directory.
- Automated new-fellow **provisioning** with a 10-step, Composio-powered playbook (Linear project, Slack channel, Google Drive workspace, Claude project context, welcome email, team announcement) with retries, an audit trail and a CI-validated tool registry.
- Owned the **compliance desk**: a 15-step case runbook covering DocuSign Connect signature verification (HMAC webhooks), KYC review, director approvals and gated stipend release; named procedure owner in the studio SOP.
- Implemented privacy-first **product analytics**: PostHog with consent gating, masked session replay and URL redaction, plus a Convex daily-active-usage model and cron-computed activation funnel in a studio analytics dashboard.
- Shipped finance and comms tooling: multi-currency (USD/QAR/EUR) stipend and compute requests with tiered approvals and receipt reconciliation, and a Resend email platform with human-approval holds, delivery tracking and a branded template builder.
- Created **PRview**, a Claude Code plugin that auto-films a narrated 1080p walkthrough of every merged PR (Claude-planned script, Playwright capture, Deepgram/Kokoro voice, Whisper captions, FFmpeg), running in GitHub Actions to publish per-PR videos and a weekly release reel.

***R&D Desk*** *— AI research workspace for venture fellows (formerly the Co-Build Hub)*

- Built per-fellow **RAG** and an "Ask" co-pilot on Convex vector search with OpenAI embeddings and tested tenant isolation, then extended it into a tool-calling **Concierge agent** (Vercel AI SDK, Claude & GPT models) with 4 specialist desks, memory, daily briefs and an eval harness.
- Built a **research engine**: scouts over arXiv, Semantic Scholar, OpenAlex, PubMed and ClinicalTrials.gov; Composio ingestion from Drive, Slack, Gmail and Linear; and a retrieval pipeline (query rewriting, reranking, chunk expansion) scoring **100% recall and 76% precision@5** on a gold eval set.
- Automated Co-Build fellow onboarding (Slack, Linear, Drive, DocuSign-gated welcome flow), added two-way **voice onboarding** with Deepgram/OpenAI speech, and generated journal-style research papers exportable to PDF and Google Docs.

**PROJECT MANAGEMENT INTERN**

APU Career Centre – Kuala Lumpur, Malaysia Jul 2024 – Nov 2024

- Developed an end-to-end survey pipeline (design, collection, cleaning, analysis) for 1,300+ students to deliver actionable insights.
- Analyzed student response data using Python, Excel, and Power BI to identify top-demanded industry certifications and programming languages, directly influencing curriculum planning and workshop development.
- Conducted market research on certification programs against current industry trends, benchmarking offerings against competitor institutions to identify gaps and opportunities.
- Contributed to performance analytics for 3+ career events, tracking engagement and satisfaction to optimize future events.
- Represented APU at NHCCE 2024 exhibition, networking with industry professionals to expand employer engagement opportunities for the career centre.

## PROJECTS

**UTOPIA OPS AGENT** May 2026

- Built a zero-dependency Node.js onboarding operations agent that turns a fellow's name and start date into an LLM-generated checklist with owners and due dates, pushes tasks to Linear, monitors completion and sends automated Slack reminders; deployed on Railway.

**QATAR MACROECONOMIC PERFORMANCE ANALYTICS** Jan 2026

- Built a 3-layer medallion data warehouse (PostgreSQL, star schema) integrating 15 macroeconomic datasets (22,500+ records) for multi-dimensional analysis.
- Identified an 82.7B QR post-World Cup GDP decline driven by Manufacturing (-14%), Mining (-6%), and Transport through sectoral trend analysis (2019–2023).
- Discovered insights on sector growth (+18% Financial Services) and developed Power BI dashboards (DAX) to visualize GDP trends, Qatarization, and regional business patterns.

**YOUTUBE ML PREDICTION PLATFORM (PREPOST ANALYTICS)** Nov 2025

- Built an ensemble ML system combining XGBoost, BERT (title sentiment and topic NLP) and CNN thumbnail analysis, trained on 51,888 videos, to predict views, engagement and CTR with 95.6% $R^2$ on test data.
- Built a full-stack analytics dashboard (React + FastAPI + SQL Server) with real-time YouTube API integration, JWT authentication, rate limiting, and scalable backend architecture.

**TRAFFIC SIGN RECOGNITION: CNN VS VISION TRANSFORMER** Mar 2026

- Built a CNN and a Vision Transformer from scratch in TensorFlow for 43 sign classes, tuned with Keras Tuner and evaluated with ROC-AUC, confusion matrices and classification reports.

**BIG FIVE PERSONALITY PREDICTOR** Aug 2025

- Trained 5 CatBoost models (one per OCEAN trait) behind a 25-question adaptive Streamlit app, with SHAP-ranked question selection and SHAP waterfall explanations for every prediction.

**DIGITAL MARKETING FUNNEL & WEB ANALYTICS** Sept 2025

- Evaluated Amazon's funnel across awareness, consideration, conversion and loyalty, and benchmarked traffic against eBay (33.9% vs 39.3% bounce rate) using Google Analytics data to recommend mobile drop-off fixes.